

LI WANG 王力

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Research Interests: Safety, Audio Deepfake Detection, Audio Language Model

EDUCATION

The Chinese University of Hong Kong , Shenzhen

Sep. 2023 –

- PhD in Computer Science
- Areas of Study: audio deepfake and anti-spoofing
- Advisor: Zhizheng Wu and Haizhou Li

Peking University (Exam-exempted postgraduate)

Sep. 2019 – Jun. 2022

- Master in Electronic and Computer Engineering (ECE)
- Areas of Study: spoken keyword spotting and speaker verification
- Advisor: Yuexian Zou

University of Electronic Science and Technology of China

Sep. 2015 – Jun. 2019

- *B.Eng.* in Information and Software Engineering
- GPA: 3.69/4.00, Ranking: 8/162(Top 4.9%)
- Core courses: Random Signal Analysis (99), Digital Signal Processing (95)

PUBLICATIONS

1. **Li Wang**, Kunyu Feng, Wan Lin, Dekun Chen, Qinke Ni, Xueyao Zhang, Lei Wang, Jie Shi, Haizhou Li, Zhizheng Wu. [PDF]
GenTraceBench: A Benchmark for Tracing Audio Deepfakes Across Pre- and Post-training Stages. *Accepted by the IEEE International Symposium on Chinese Spoken Language Processing (ISCSLP 2026)*.
2. **Li Wang***, Jindong Wang*, Wan Lin*, Kunyu Feng*, Lei Wang, Rizhao Cai, Ce Fang, Jinzhe Xue, Jie Shi, Haizhou Li, Zhizheng Wu. [Project] [Code]
RealComm: Benchmarking and Adapting Audio Deepfake Detection over Real Communication Channels. *Submitted to IEEE/ACM Transactions on Audio, Speech, and Language Processing (TASLP)*. (* Equal contribution.)
3. Wan Lin, **Li Wang**, Jindong Wang, Kunyu Feng, Zhizheng Wu. [PDF] [Project]
Teffic-Audio: Tell Fact from Fiction. *Technical Report, arXiv:2607.28351, 2026*.
4. Yuxiang Wang, Hongyu Liu, Yijiang Xu, Qinke Ni, **Li Wang**, Wan Lin, Kunyu Feng, Dekun Chen, Xu Tan, Lei Wang, Jie Shi, Zhizheng Wu. [PDF] [Project] [Code]
VoxSafeBench: Not Just What Is Said, but Who, How, and Where. *Preprint, arXiv:2604.14548, 2026*.
5. Yupei Li*, **Li Wang***, Yuxiang Wang, Lei Wang, Rizhao Cai, Jie Shi, Björn W. Schuller, Zhizheng Wu. [PDF]
DFALLM: Achieving Generalizable Multitask Deepfake Detection by Optimizing Audio LLM Components. *Technical Report, arXiv:2512.08403, 2025*. (* Equal contribution.)
6. Xueyao Zhang, Chaoren Wang, Huan Liao, Ziniu Li, Yuancheng Wang, **Li Wang**, Dongya Jia, Yuanzhe Chen, Xiulin Li, Zhuo Chen, Zhizheng Wu. [PDF]
SpeechJudge: Towards Human-Level Judgment for Speech Naturalness. *Proceedings of the 14th International Conference on Learning Representations (ICLR 2026)*.
7. **Li Wang**, Xiao Lei, Haorui He, Lei Wang, Jie Shi, Zhizheng Wu. [PDF]
Over-the-Air Adversarial Attacks and Detection for Automatic Speaker Verification. *IEEE/ACM Transactions on Audio, Speech, and Language Processing (TASLP)*, 2026.
8. Peizhuo Liu*, **Li Wang***, Renqiang He*, Haorui He, Lei Wang, Huadi Zheng, Jie Shi, Tong Xiao, Zhizheng Wu. [PDF]
SpMis: An Investigation of Synthetic Spoken Misinformation Detection. *Proceedings of the IEEE Spoken Language Technology Workshop (SLT 2024)*. (**Best Paper Finalist, Top 2.5%**)

9. **Li Wang**, Jiaqi Li, Yuhao Luo, Jiahao Zheng, Lei Wang, Hao Li, Ke Xu, Chengfang Fang, Jie Shi, Zhizheng Wu. [PDF]
AdvSV: An Over-the-Air Adversarial Attack Dataset for Speaker Verification. *Proceedings of the 49th IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP 2024)*.
10. Jiaqi Li, **Li Wang**, Liumeng Xue, Lei Wang, Zhizheng Wu. [PDF]
An Initial Investigation of Neural Replay Simulator for Over-the-Air Adversarial Perturbations to Automatic Speaker Verification. *Proceedings of the 49th IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP 2024)*.
11. **Li Wang**, Rongzhi Gu, Weiji Zhuang, Peng Gao, Yujun Wang, Yuexian Zou. [PDF]
Learning Decoupling Features Through Orthogonality Regularization. *Proceedings of the 47th IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP 2022)*.
12. **Li Wang**, Rongzhi Gu, Nuo Chen, Yuexian Zou. [PDF]
Text Anchor Based Metric Learning for Small-Footprint Keyword Spotting. *Proceedings of the 22nd Annual Conference of the International Speech Communication Association (Interspeech 2021)*.

EXPERIENCE

Shenzhen Research Institute of Big Data (SRIBD) Nov. 2022 – Aug. 2023

Research Assistant, Institute of Basic Theory and Algorithm of Big Data, Shenzhen, China

Leading a project with Huawei on detecting physical attacks in speaker verification technology.

- Successfully generated a dataset of 1.1 million data samples for physical attacks on speaker verification.
- Determined the technical route for the defense method, which was fully affirmed and supported by Huawei.
- Contributed to developing a reliable defense mechanism for speaker verification against physical attacks.
- Demonstrated strong leadership, problem-solving, and technical skills throughout the project.

International Digital Economy Academy (IDEA) Jul. 2022 – Oct. 2022

Financial Machine Learning Engineer, AI Finance and Deep Learning Research Center, Shenzhen, China

Create financial factors (features) and design deep learning models to predict stock returns.

INTERNSHIPS

AILAB, ByteDance Jul. 2021 – Mar. 2022

- Topic: Audio fingerprinting and music segmentation
- Job Description: Proposed the audio fingerprint temporal localization algorithm. The accuracy rate was improved from 95.80% to 97.90%, and the IOU of temporal localization was 0.62.

Audio Engineering Department, HuaWei MTI Apr. 2021 – Jun. 2021

- Topic: Music separation
- Job Description: Replicated D3Net and trained and tested on the MUSDB dataset, separating vocals, drums, bass and other tracks separately, with a test average SDR of 6.43.

AI Lab, Tencent May. 2020 – Nov. 2020

- Topic: Speech separation and speaker verification
- Job Description: Testing the performance of the speech separation model GALR on a speaker verification task, collating speaker information from 2,000,000 pieces of audio data.

PROJECTS

Huawei HarmonyOS 7 Audio Deepfake Detection Jan. 2025 – Present

Contributor Industry-academia collaboration with Huawei

- Contributed to the research, development, and product deployment of the audio deepfake detection capability released in Huawei HarmonyOS 7.
- Developed generalizable and robust detection techniques for diverse speech synthesis methods and distortions in real communication channels.

Research on Voice Wake-up Technology of Artificial Intelligence

Jan. 2021 – Jan. 2022

Leader Collaborative project between the lab and Xiaomi during graduate school.

- Communicated with Xiaomi to track project progress, understand data requirements.
- Optimized network design and data enhancement to improve model performance.
- Reduced the false rejection rate from 20% to 2% and achieved an accuracy rate of 99.7%.
- Collaborated on personalized wake-up task and submitted ICASSP2022 paper as first author.

SERVICES

Reviewer

- IEEE/ACM Transactions on Audio, Speech, and Language Processing (IF 4.1)
- IEEE International Conference on Acoustics, Speech and Signal Processing
- IEEE Signal Processing Letters (IF 3.201)
- National Conference on Man-Machine Speech Communication (NCMMSC)
- International Conference on Asian Language Processing (IALP)
- Computer Speech & Language (IF 3.252)
- Workshop on Automatic speech Recognition and Understanding

Standards Development

- Contributed to the development of “Method for Identifying AI-Generated Synthetic Content” (Metadata Implicit Identification for Text Files), published by National Technical Committee 260 on Cybersecurity of SAC (TC260).

Student Volunteer

- IEEE Spoken Language Technology Workshop 2024

Teaching Assistant

- 2023 Fall, CSC1001 Introduction to Computer Science: Programming Methodology, CUHK-Shenzhen.
- 2024 Spring, CSC3160/AIR6063 Fundamentals of Speech and Language Processing/Spoken Language Processing, CUHK-Shenzhen.
- 2024 Fall, DDA3020 Machine Learning, CUHK-Shenzhen.
- 2025 Spring, DDA3020 Machine Learning, CUHK-Shenzhen.

HONORS AND AWARDS

1. National Second Prize of National University Student Information Security Competition 全国大学生信息安全竞赛 国家级二等奖 (2018)
2. Undergraduate People's First Class Scholarship 本科生人民一等奖学金 (2015)
3. Undergraduate People's First Class Scholarship 本科生人民一等奖学金 (2016)
4. Undergraduate People's First Class Scholarship 本科生人民一等奖学金 (2017)
5. Outstanding Volunteer of the 9th Paralympic Games of the People's Republic of China. 中华人民共和国第九届残疾人运动会优秀志愿者 (2015)
6. Peking University 4x100m Runner-up in the School Games. 北京大学校运会 4x100 米亚军 (2021)

SKILLS

- Erhu Grade 10.
- Programming Languages: proficient in Python, familiar with Java, and know about C++.
- Deep Learning Frameworks: proficient in PyTorch.